

Table 1 Fragile X mouse synaptic plasticity phenotypes

Category	Region	Fragile X Mouse Phenotype	Age	References
mGluR LTD	hippocampus	enhanced	P25-30	Huber et al., 2002; Hou et al., 2006; Bhattacharya et al., 2012; Michalon et al., 2012
mGluR LTD	hippocampus	does not require new protein synthesis	4-12 wk	Nosyreva and Huber, 2006; Hou, et al., 2006; Zang et al., 2009**
mGluR LTD	hippocampus	*enhanced and not PS-dependent	P35-42	Iliff et al., 2012
mGluR LTD	cerebellum	enhanced	3-7 wk	Koekkoek et al., 2005
mGluR LTD	hippocampus	enhanced and does not require new protein synthesis	3-7 wk	Volk et al., <i>J Neurosci</i> , 2007
LTP	hippocampus	NONE	20-26 wk; 8-10 wk; 3-12 month	Godfraind et al., 1996; Li et al., 2002; Larson et al., 2005
L-LTP	hippocampus	NONE	5-7 wk; 2-4 month	Paradee et al., 1999; Zhang et al., <i>J</i> 2009
LTP	hippocampus	deficient	2 wk; 6-8 wk	Hu et al., 2008; Shang et al., 2009
LTP	hippocampus	deficient with weak stimulus; normal with strong stimulus	2-3 month	Lauterborn et al., 2007
LTP	hippocampus	enhanced B-adrenergic-facilitated heterosynaptic LTP (PS-dependent)	3-4 month	Connor et al., 2011
LTP priming	hippocampus	does not require new protein synthesis (mGluR-dependent)	6-10 wk	Auerbach and Bear, 2010
LTP	anterior cingulate ctx	deficient	6-8 wk	Zhao et al., 2005; Xu et al., 2012
LTP	anterior cingulate ctx	impaired facilitation of LTP by 5-HT _{2A} R agonist	6-8 wk	Xu et al., 2012
LTP	somatosensory, temporal ctx	deficient	8-10 wk; 3 month	Li et al., 2002; Hayashi et al., 2007
LTP	somatosensory ctx	delayed window for plasticity	P3-10	Harlow et al., 2010
LTP	visual ctx	deficient (mGluR-dependent)	P13-25	Wilson and Cox, 2007
LTP	anterior piriform ctx	deficient in aged mice; normal in 3-6 mo mice	6-18 month	Larson et al., 2005
LTP	amygdala	impaired (mGluR-dependent)	6-8 wk; 3.5-6 mo	Zhao et al., 2005; Suvrathan et al., 2010
STD-LTP	somatosensory ctx	deficient with weak stimulus	P10-18	Desai et al., 2006
STD-LTP	prefrontal ctx	deficient with weak stimulus; normal with strong stimulus	P14-23	Meredith et al., 2007
homeostasis	hippocampus	deficient translation-dependent scaling	P6-7 slice culture	Soden and Chen, 2010
homeostasis	hippocampus	normal transcription-dependent scaling	P6-7 slice culture	Soden and Chen, 2010
experience-dependent	visual ctx (in vivo)	altered ocular dominance plasticity	LTD	Dolen et al., 2007
experience-dependent	somatosensory ctx	deficient experience-dependent plasticity (induced by whisker trimming)	LTD	Bureau et al., 2008

LEGEND: Fragile X mouse models have multiple altered forms of synaptic plasticity across multiple brain regions. The majority of phenotypes were assessed in the *Fmr1* KO mouse which lacks FMRP. *Assessed using a CGG knock-in mouse which models FX premutation. **Assessed using an FMRP point mutant mouse with disrupted FMRP-mRNA binding.

LTD (Figure 1B). However, while the magnitude of L-LTP is unchanged, it is possible that L-LTP is qualitatively different in its requirement for new protein synthesis when FMRP is absent, as is the case for mGluR-LTD (and LTP priming, see below). Therefore, it will be important to test the protein synthesis-dependency of L-LTP in *Fmr1* KO mice to show that FMRP truly does not play a role in regulating the persistence of LTP.

Alternatively, FMRP may be required for the regulation of local but not somatic translation in the context of L-LTP (Figure 1C). L-LTP is traditionally induced by multiple trains of high frequency tetanus or theta burst stimulation, protocols that rely on cell-wide transcription and translation [63-65]. L-LTP was characterized in the *Fmr1* KO mouse using these classical paradigms [61,62]. However, using a less intense induction protocol